

Cruz Castillo

Email: ccastillo49@gatech.edu • [Website](#)

CONTACT INFORMATION	Department of Mathematics Skiles. 686 Cherry St. N.W.	Georgia Institute of Technology Atlanta, GA 30332
EDUCATION	University of Illinois at Urbana-Champaign Ph.D. in Mathematics Advisor: Scott Ahlgren M.S. in Mathematics B.S. in Mathematics with Highest Distinction	August 2021-May 2026 December 2021 May 2020
PUBLICATIONS	Non-vanishing for quartic Hecke L-functions and ranks of elliptic curves. 65pp. With Alexandre de Faveri and Alex Dunn	Submitted(2026)
	Arithmetic properties of generalized Frobenius partitions. 16pp. With Scott Ahlgren	Submitted(2025)
	Eisenstein series modulo prime powers. 12pp. Accepted for publication at Proc. of the AMS(2025) With Scott Ahlgren and Clayton Williams	
	Surfaces associated to zeros of automorphic L-functions. 48pp. With Debmalya Basak and Alexandru Zaharescu.	Submitted(2024)
	Sign changes of the error term of the Piltz divisor problem 29pp. Int. Math. Res. Not. IMRN 2024, no.20,13191-13220. With Siegfred Baluyot	
SCHOLARSHIPS, FELLOWSHIPS, AND AWARDS	NSF Graduate Research Fellowship Sloan Scholar Alfred P. Sloan Foundation's Minority Ph.D. Program Graduate College Master's Fellowship University of Illinois at Urbana-Champaign Math Alliance Scholar National Alliance for Doctoral Studies in the Mathematical Sciences President's Award Program Honors University of Illinois at Urbana-Champaign	2022- 2026 2021-2026 2020-2021 2019-2026 2017-2020
PRESENTATIONS [†] = upcoming	<i>Non-vanishing for quartic Hecke L-functions and ranks of elliptic curves</i> [†] AMS Sectional: Recent Developments in Number Theory Kennesaw State University GA <i>Non-vanishing for quartic Hecke L-functions and ranks of elliptic curves</i> The 2026 Gainesville International Number Theory Conference Gainesville, FL <i>Surfaces associated to zeros of automorphic L-functions</i> JMM: Research Presentations by Math Alliance Doctorates	October 2026 March 2026 January 2026

	JMM: Recent Trends in Analytic Number Theory Washington D.C.	
	<i>Surfaces associated to zeros of automorphic L-functions</i> Georgia Tech Number Theory Seminar	September 2025
	<i>Arithmetic properties of m-colored Frobenius partitions</i> 37 th Automorphic Forms Workshop	April 2025 University of North Texas
	<i>Surfaces associated to zeros of automorphic L-functions</i> UIUC Number Theory Seminar	October 2024
	<i>Surfaces associated to zeros of automorphic L-functions</i> AMS Sectional: Special Session on Additive Number Theory and Modular Forms University of Texas at San Antonio	September 2024
	<i>A surface associated to the zeros of the Riemann zeta function</i> Princeton/IAS Number Theory Seminar	February 2024 Joint with Alexandru Zaharescu
	Sign changes of the error term of the Piltz divisor problem CRG Weekly-Seminar	September 2023 University of British Columbia
	Sign changes of the error term of the Piltz divisor problem UIUC Number Theory Seminar	May 2023
POSTER PRESENTATIONS	<i>Surfaces associated to zeros of automorphic L-functions</i> Midwest Arithmetic Geometry and Number Theory Conference	March 2025 Ohio State University
	<i>Sign changes of the error term of the Piltz divisor problem</i> Midwest Arithmetic Geometry and Number Theory Conference	October 2023
	<i>Sign changes of the error term of the Piltz divisor problem</i> Inclusive Paths in Explicit Number Theory Summer School	July 2023
	<i>Daphnia population dynamics with varying genotypes, two food sources, and parasites</i> University of Illinois Housing: Researcher's Initiative	May 2018
OUTREACH	Modular Forms: The Fifth Fundamental Operation Lake Forest College Math Seminar	October 2023
	Explicit Formulas Related to the Riemann ζ function Rose-Hulman Institute of Technology Math Seminar	October 2023 Interactive Website
	Teaching assistant at Spartan Academy Marian Catholic High School Led STEM activities on robotics, coding, and engineering for K-8 students.	Summer 2018 Chicago Heights, IL
TEACHING EXPERIENCE	Georgia Institute of Technology Summer Edge Program Instructor: Analysis	Summer 2026
	University of Illinois at Urbana-Champaign Teaching Assistant: Calculus 3 (NetMath program)	Summer 2025

Teaching Assistant: Calculus 3 Fall 2024-Spring 2025
Ranked “Excellent” by students: Spring 2025

Teaching Assistant: Calculus 1 Fall 2021-Spring 2022
Ranked “Excellent” by students both semesters.

Course Assistant: CS 101 (Introductory Course on Python) 2018-2020

CONFERENCES AND WORKSHOPS ATTENDED 37th **Automorphic Forms Workshop**
 University of North Texas April 2025

Midwest Arithmetic Geometry and Number Theory Conference
 Ohio State University March 2025

AMS Sectional: Special Session on Additive Number Theory and Modular Forms
 University of Texas at San Antonio September 2024

Comparative Prime Number Theory Symposium (PIMS) June 2024
 University of British Columbia, Vancouver

Clifford Lectures: The Web of Modularity February 2024
 Tulane University

Midwest Arithmetic Geometry and Number Theory Conference
 University of Michigan October 2023

Institute on Teaching and Mentoring October 2023
 Southern Regional Education Board

Inclusive Paths in Explicit Number Theory Summer School July 2023
 BIRS at UBC-Okanagan, Kelowna, BC, Canada

Midwest Arithmetic Geometry and Number Theory Conference
 University of Illinois at Chicago October 2022

BB 5th EU/US Summer School and Workshop on Automorphic Forms and Related Topics August 2022
 University of Sarajevo

Thematic Program in p-adic L-functions and Eigenvarieties July 2022
 University of Notre Dame

NSF/CBMS Research Conference Ramanujan’s Ranks, Mock Theta Functions and Beyond May 2022
 University of Texas Rio Grande Valley

Field of Dreams Conference November 2019
 Math Alliance, Purdue University
 St. Louis, Missouri

Career Paths in the Mathematical Sciences: An IMA/Math Alliance Workshop June 2019

IM&A, University of Minnesota

SERVICE

Co-organizer

[Palmetto Number Theory Series \(PANTS\) XLI](#)

Georgia Tech

October 3-4, 2026

Co-organizer

Fall 2026- Georgia Tech Number Theory Seminar

2025-Spring 2026
Climate, Equity and Inclusion Committee

UIUC

Graduate Student Mentor

Illinois Mathematics Lab

Project title: Automated Proofs of Partition Congruences

Students: Hao Cai, Tyler Cushing, Jennifer Rotter, Eeshan Zele

Fall 2023

UIUC

Committee Member

Committee to prepare surveys on Math Alliance Scholars

Fall 2022

Math Alliance

Co-organizer

UIUC Graduate Number Theory Seminar

Fall 2022-Spring 2026

Peer Mentor

Peer Mentoring in a Virtual University Setting Program

University of Illinois at Urbana-Champaign

Fall 2021-Spring 2023